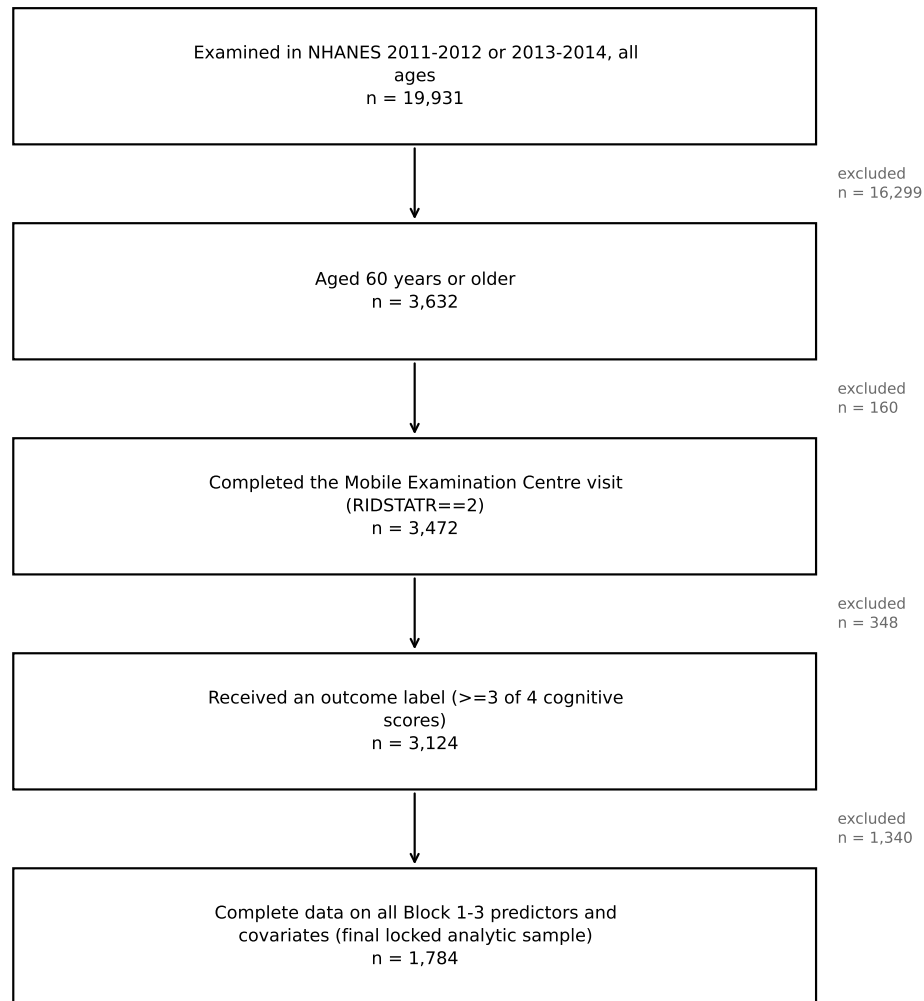


## Figures

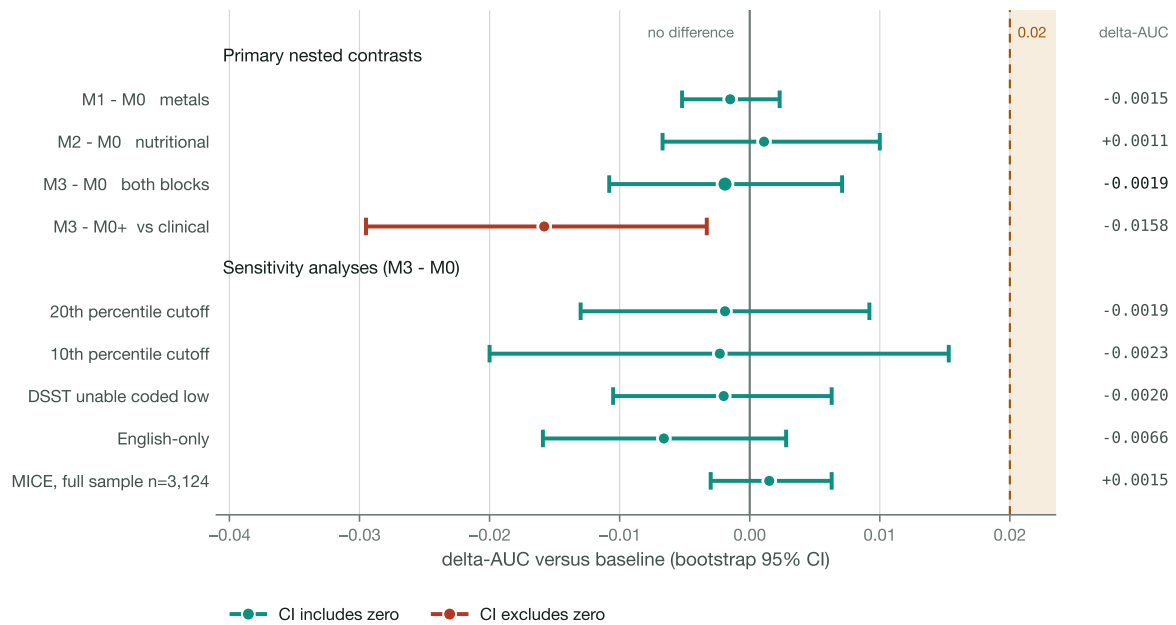
Reported biomarker–cognition associations do not translate into predictive utility: a pre-registered analysis of NHANES 2011–2014

Shreyan Kancharla, Neil Patel

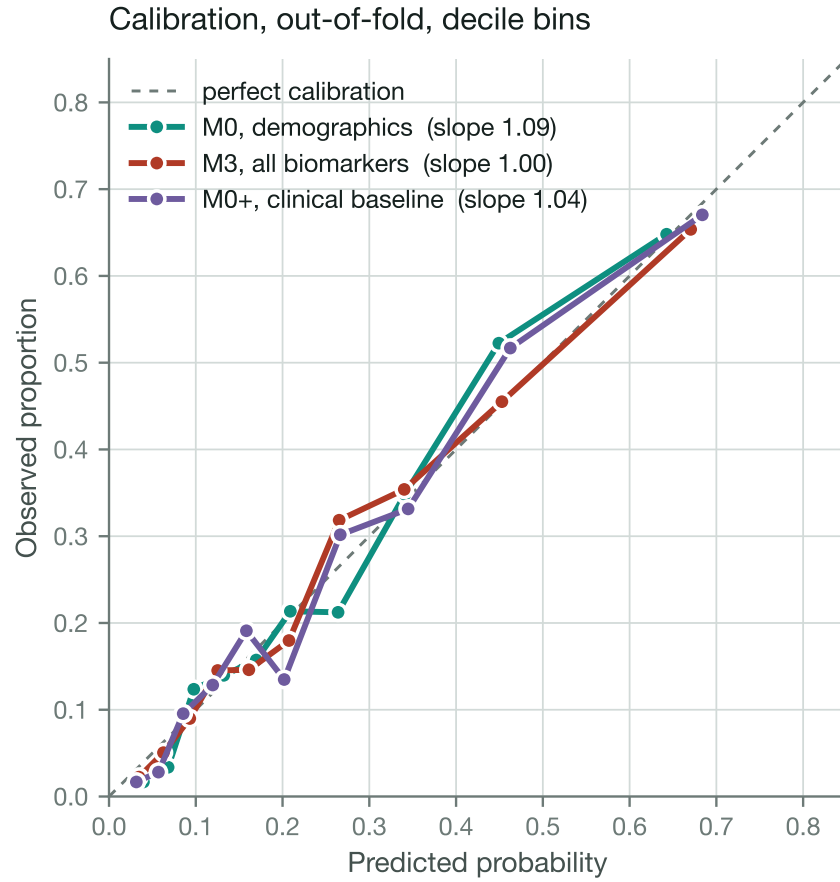
Participant flow, NHANES 2011-2012 and 2013-2014



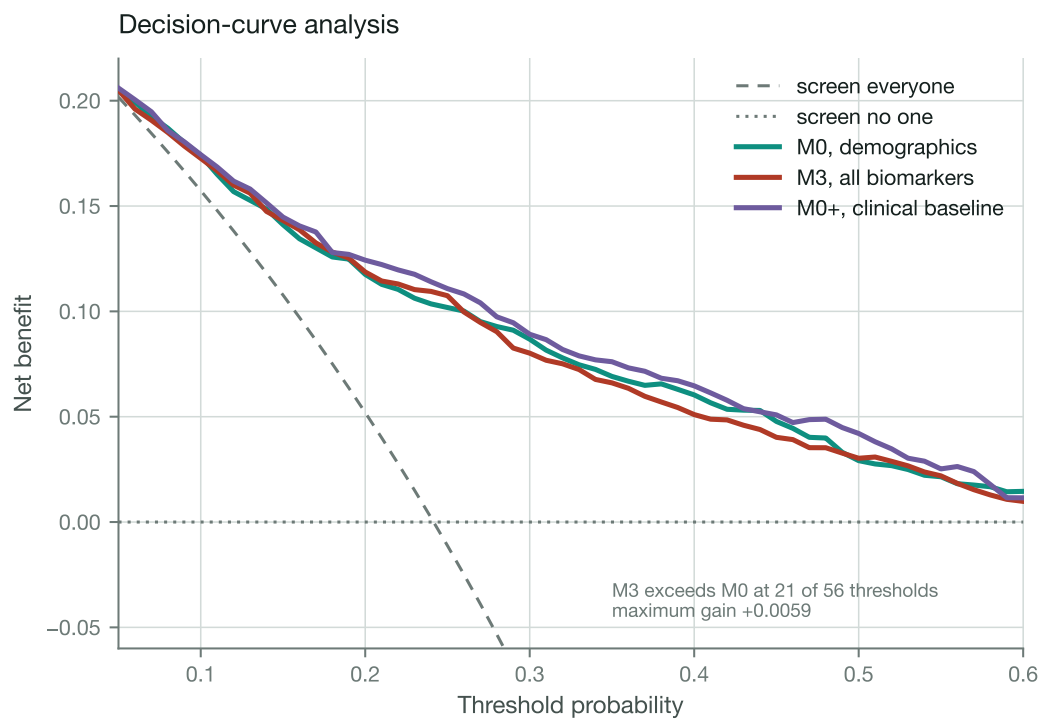
**Figure 1:** Participant flow from NHANES 2011–2012 and 2013–2014 examined participants to the final locked analytic sample. Reasons for exclusion are given at each stage; the final step’s exclusions (missing predictor data) are not mutually exclusive across blocks.



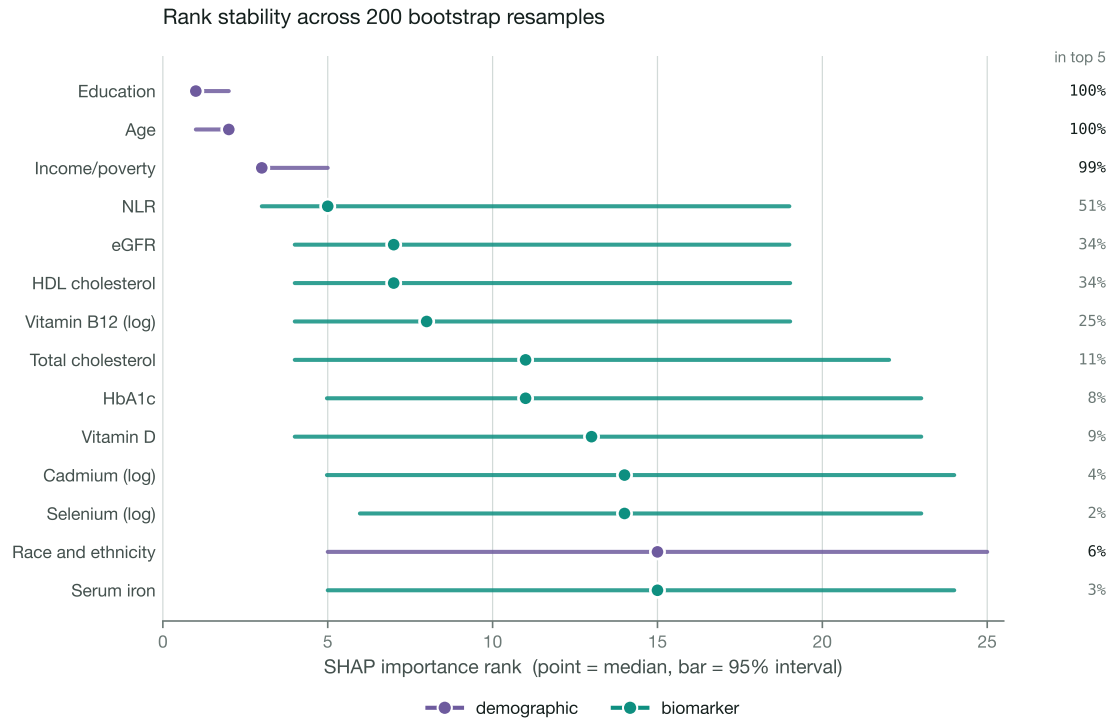
**Figure 2:** Forest plot of  $\Delta$ AUC contrasts with bootstrap 95% confidence intervals: the four primary nested contrasts (M1–M0, M2–M0, M3–M0, M3–M0+) and the six pre-specified sensitivity analyses for M3–M0. Red markers indicate intervals excluding zero. The shaded band marks the pre-specified 0.02 meaningful-difference threshold.



**Figure 3:** Calibration, out-of-fold, by decile of predicted probability, for M0 (demographics), M3 (all biomarkers) and M0+ (clinical baseline). Calibration slope for each model is given in the legend; the dashed line is perfect calibration.

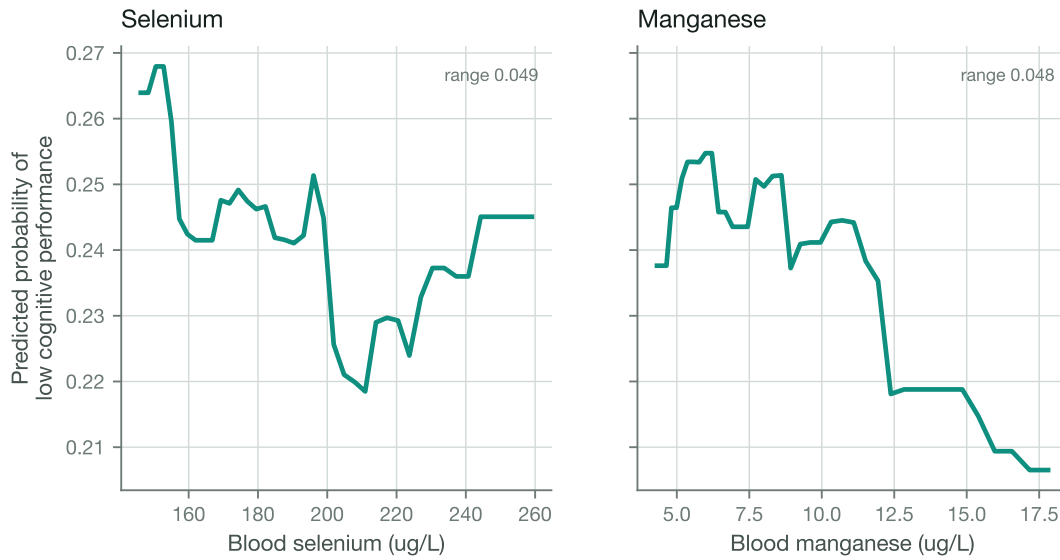


**Figure 4:** Decision-curve analysis: net benefit against threshold probability for M0, M3 and M0+, against the treat-all and treat-none reference strategies.



**Figure 5:** SHAP feature-importance rank stability across 200 bootstrap resamples of the gradient-boosted M3 model. Points are median rank, bars are the 95% interval; the right-hand column gives the percentage of resamples in which each feature appeared in the top 5. Demographic predictors (purple) are essentially deterministic in rank; biomarkers (teal) are not.

No U-shaped exposure-response is present for either essential metal



**Figure 6:** Partial dependence of predicted probability of low cognitive performance on blood selenium and blood manganese concentration (gradient-boosted M3 model), the two exposures for which a non-monotonic (U-shaped) exposure-response was anticipated in advance. Neither shows one; the predicted probability varies by approximately 0.05 across each exposure's observed range.